

# Research on Data-driven Deep Learning Models for Tennis Matches and Their Performance Optimization

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**Abstract**—This paper proposes an innovative deep learning model based on the combination of GCN and LSTM for predicting the outcome of tennis matches. This paper first constructs a dynamic relationship graph in the game, converts factors such as player interactions, scoring, and field status in the game data into nodes and edges of the graph, and captures the relationships and interactions between players through the GCN model. Then, the LSTM network is used to model the time series characteristics of the game to help analyze the dynamically changing game status during the game. Compared with traditional machine learning models, the model proposed in this paper can more effectively handle complex spatiotemporal dependencies, thereby improving prediction accuracy. A set of real tennis matches from different seasons was used to evaluate the model by means of precision, accuracy, recall, and F1. Experiments show that the prediction precision of this model is up to 90%, which is about 12% better than the conventional SVM and decision trees. Moreover, the optimized model has obvious superiority in accuracy and recall, which proves the validity of FNN and LSTM in the prediction of sports events.

**Keywords**—Tennis match, deep learning, graph neural network, LSTM, data prediction, performance optimization

## I. INTRODUCTION

In recent years, sports match prediction has become a hot research direction in the field of artificial intelligence. With the rapid development of data analysis technology, especially driven by big data and deep learning methods, the prediction model of sports events has gradually evolved from traditional statistical analysis to more complex machine learning and deep learning methods. These methods can not only process historical data, but also analyze real-time data and make accurate predictions. Especially in the field of tennis match prediction, data-driven analysis methods are increasingly widely used. Predicting tennis matches is a very challenging task, mainly because the results of the game are affected by many factors [1]. The player's technical level, physical condition, psychological state, and the game environment (such as venue type, weather, etc.) may have an important impact on the progress and final results of the game. In addition, the dynamic changes of the game also make the prediction process more complicated [2]. Despite this, tennis match prediction is still an important topic worthy of research, and its application prospects are broad, including in many fields such as event analysis, sports betting, and player training.

Tennis match data is the basis for accurate prediction, mainly from historical game records, real-time game data, and the health and physical condition of players [3].

Historical match data covers details such as players' game scores, scores, mistakes, serves and receives, and provides players' performances under different opponents, venues and conditions. These data reflect players' competitive status, skill characteristics and game strategies, which are crucial for predicting game results [4]. The collection of real-time game data relies on modern technical means, including sensors, cameras and other data collection equipment. During the game, real-time data can provide instant feedback to the prediction model and help adjust the prediction results [5]. The health and physical condition of the players also play a decisive role in the outcome of the game, especially in long-term games. The physical decline of the players may cause their performance to decline sharply, thus affecting the results of the game.

Moreover, the game data is very diverse. For example, the scoring record, the movement trajectory of the player, the speed of the ball, the type of court, etc., can influence the progress of the match. Because of the large quantity of non-linear relations and the complexity of the model, it is difficult to deal with this kind of data by conventional statistical methods [6]. Therefore, it is especially important to deal with such complicated relationships with deep learning models. Deep learning models such as CNN and RNN have been shown to be effective in capturing complex patterns and non-linear relationships.

The innovative point of this research is that the combination of GCN and LSTM can increase the precision and stability of tennis game forecast. Traditional CNN, RNN and so on have made some achievements in dealing with time series data, but there are still some limitations when dealing with structured data [7]. This paper introduces GCN into the tennis match prediction model to better capture the complex relationships between players and between players and the court. GCN can model the interaction between players through graph structures, thereby improving the model's ability to understand the complex dynamics of the game. On the other hand, LSTM, as a recurrent neural network that processes time series data, can effectively capture temporal dependencies [8]. In a tennis match, the game process is a dynamic time series data. LSTM can model the changes that occur in the game through a memory mechanism and predict the subsequent results of the game. By combining GCN and LSTM, this study aims to achieve accurate prediction of the results of tennis matches and be able to make real-time dynamic adjustments to the progress of the game.

## II. DEEP LEARNING MODEL DESIGN

### A. Model architecture

The prediction task of tennis matches is a typical multimodal data analysis problem, involving information in multiple dimensions such as historical match data, player status, match environment, and real-time data. Traditional machine learning methods often have difficulty processing such complex and multi-level information, especially during the game, the interaction between players, changes in match scores, and the influence of factors such as venue conditions cannot be ignored [9]. A new GCN and LSTM model is proposed to improve the precision and stability of tennis game forecast. GCN can effectively model graph structure data and is suitable for modeling relationships between players. In tennis matches, the interaction between players not only affects the direction of the game, but also determines the final result of the game. Therefore, modeling the player relationship in the game through a graph structure can better capture this complex interactive information. LSTM is suitable for capturing time series features [10]. During the game, the changes in the player's status, score, etc. change dynamically over time. Therefore, analyzing the evolution of time series data in the game through LSTM can provide a more accurate prediction basis for the prediction model. The graph convolution operation can aggregate the neighborhood information of the node. The calculation method of graph convolution in GCN is:

$$h_i^{(k+1)} = \sigma(\sum_{j \in \mathcal{N}(i)} W^{(k)} h_j^{(k)} + b^{(k)}) \quad (1)$$

$h_i^{(k)}$  represents the hidden state of node  $i$  in the  $k$  layer,  $\mathcal{N}(i)$  represents the neighborhood of node  $i$ ,  $W^{(k)}$  is the weight matrix of the  $k$  layer,  $b^{(k)}$  is the bias term, and  $\sigma$  is the activation function. LSTM captures long-term dependencies in time series through a gating mechanism. At each time step  $t$ , the state update of LSTM can be expressed by the following formula:

$$c_t = f_t * c_{t-1} + i_t * \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (2)$$

$i_t, f_t, o_t$  are the input gate, forget gate and output gate respectively,  $W, U, b$  are the weight matrix and bias term of each layer.

### B. Combined model based on GCN and LSTM

#### 1) Constructing dynamic relationship diagram in the game

The game data can be represented as a weighted graph, in which the nodes represent the players, the edges represent the relationships among the players, and the values of the edges are calculated based on information such as the difference in the score and the interaction in the game [11]. Through graph convolution operations, this paper can aggregate node neighbor information to effectively capture the dynamic changes of the game. This paper can define the adjacency matrix  $A$  of the weighted graph based on the difference in player scores:

$$A_{ij} = \frac{|s_i - s_j|}{\max(s_i, s_j)} \quad (3)$$

$s_i$  and  $s_j$  represent the scores of players  $i$  and player  $j$  at

the current time point, respectively.

#### 2) Design of time series modeling

A tennis match is a typical time series process, in which the scores, physical fitness, emotional state and other factors in the match change over time. LSTM can effectively capture these time series features through its unique gating mechanism. LSTM is used to model each time step in the match. For each game, the change of the player's state will affect the progress of the entire game. The LSTM can keep the past history information through its storage unit at every time step. It can control the flow of information by means of input gate, forget gate and output gate.

For each moment  $t$ , the input of LSTM is determined by the current game state  $x_t$  and the state  $h_{t-1}$  of the previous moment:

$$x_t = [s_t, f_t, c_t] \quad (4)$$

$s_t$  is the score at the current time point,  $f_t$  is the physical condition of the player at the current time point, and  $c_t$  is the emotional or psychological state of the player.

#### 3) Training process and information transfer

In the process of combining GCN and LSTM, the model obtains the interaction information between players. This information will then be used as the input of LSTM to capture the dynamic characteristics of the game that change over time. The output of GCN  $h_i^{(k+1)}$  will be passed to the input layer of LSTM as part of the time series data:

$$h_{t+1}^{(LSTM)} = \text{LSTM}(h_i^{(k+1)}, h_{t-1}^{(LSTM)}) \quad (5)$$

### C. Optimization goals

The main optimization goal of this study is to improve the accuracy of tennis match predictions and the generalization ability of the model. In deep learning models, accuracy and generalization ability are usually opposing goals, so this paper needs to find a suitable balance between the two.

#### 1) Improving prediction accuracy

To increase the precision of forecast, the thesis begins with the adjustment of network structure. By adding more GCN and LSTM layers, it is possible to better capture complex non-linear relations. Secondly, people can improve the stability of training by optimizing the activation function, adjusting the learning rate, and so on. The model loss function is the kernel of optimization. WMSE is used as loss function in this paper:

$$\mathcal{L} = \sum_{t=1}^T w_t \cdot (y_t - \hat{y}_t)^2 \quad (6)$$

$y_t$  is the actual value,  $\hat{y}_t$  is the predicted value,  $w_t$  is the weight of time step  $t$ , and  $T$  is the number of time steps.

#### 2) Enhanced generalization ability

To improve the generalization capability of the model, people use the early stopping and regularizing technology. The early stop method monitors the performance of the model on the validation set, and ends the training as soon as the validation error does not decrease [12]. Regularizers like L2 regularizes can effectively avoid overfitting, and thus enhance the generalization capability of the model. L2

regularization avoids overfitting by adding a weight penalty term to the loss function:

$$\mathcal{L}_{\text{reg}} = \mathcal{L} + \lambda \sum_i W_i^2 \quad (7)$$

$W_i$  is the weight of the model, and  $\lambda$  is the regularization coefficient.

### 3) Model training process

The model training process includes the following steps: First, the GCN module models the relationship between players and generates interaction information between players; then, the LSTM module models the game process based on time series data; finally, the model parameters are optimized through the error back propagation algorithm. Back propagation updates the model parameters through the gradient descent method:

$$\theta = \theta - \eta \cdot \nabla_{\theta} \mathcal{L} \quad (8)$$

Among them,  $\theta$  represents the model parameters,  $\eta$  is the learning rate, and  $\nabla_{\theta} \mathcal{L}$  is the gradient of the loss function.

## III. PERFORMANCE OPTIMIZATION

### A. Model parameter adjustment

The performance of deep learning models is highly dependent on the selection of hyperparameters, especially for complex prediction tasks. The hyperparameter adjustment of the model is very important for the prediction of tennis matches. Common hyperparameters include learning rate, batch size, and network level. The adjustment of each hyperparameter can significantly affect the training efficiency and the forecast precision.

Learning rate is a key parameter in optimization. Learning rate determines the step size of adjusting model parameters in every update. Too high learning rate will make the model skip the optimum solution, and the lower learning rate will make the training time too long or even stop at the local optimum solution. Therefore, reasonable adjustment of learning rate is very important. To optimize the selection of learning rate, people adopt a learning rate decline strategy, i. e., the learning rate decreases gradually with the progress of the training. The learning rate decay formula is as follows:

$$\eta_t = \eta_0 \cdot \frac{1}{1 + \alpha \cdot t} \quad (9)$$

$\eta_0$  is the initial learning rate,  $\alpha$  is the decay factor,  $t$  is the number of training steps, and  $\eta_t$  is the current learning rate.

Batch size is the number of samples that are used each time a model parameter is updated. Selecting a suitable batch size helps to speed up the training process and prevent excessive training data from being overloaded. Larger batches are generally more stable in updating parameters, but may lead to slower training. Smaller batches can speed up training, but may result in more noise in the training process. Therefore, the proper selection of batch size is the key to the optimization of training efficiency. The number of network layers determines the depth of neural networks. Deep neural networks (DNNs) have been shown to be able to capture more complex data features, but they can lead to gradient vanishing or explosion. Therefore, controlling the number of network layers and avoiding too deep networks are important

strategies to prevent model instability.

### B. Application of data augmentation technology

In tennis match prediction, data diversity is crucial to model training. Tennis match data often has high time-varying and individual differences. A single data set may cause the model to overfit to a specific scenario, thus affecting its generalization ability. Therefore, data augmentation technology plays a key role in such tasks.

The objective of data augmentation is to expand or transform existing data to produce more training samples. Data augmentation can be performed in several ways for tennis match data. For example, it is possible to increase the variety of data by simulating different matches, generating synthetic match data, and adjusting the random fluctuation of players' performance, so as to enhance the robustness and precision of the model. Assuming that the original data set is  $D = \{d_1, d_2, \dots, d_n\}$ , a new data set  $D'$  is generated through the augmentation operation, then the augmented data set is:

$$D' = \{d_1, d_2, \dots, d_n, \tilde{d}_1, \tilde{d}_2, \dots, \tilde{d}_m\} \quad (10)$$

Among them,  $\tilde{d}_i$  is the new sample generated by the data enhancement operation, and  $m$  is the number of enhanced samples generated. Through such enhancement operations, the model can learn a wider range of game scenarios, thereby improving its generalization ability.

### C. Solution to the overfitting problem

Overfitting is one of the most common problems in the training of deep learning models, especially when the training data is small. It is possible that the model will learn the noise from the data rather than the actual rule, which will influence the forecast performance on the new data. In response to the overfitting problem in tennis match prediction, this article can adopt several effective solutions, such as regularization, Dropout technology and early stopping strategy.

Dropout is a technology that randomly "discards" some neurons during training. The purpose is to improve the generalization ability of the model by reducing the dependency between neurons. Dropout usually randomly selects some neurons with a certain probability to have zero output during training to prevent some neurons from having too much influence on the prediction results of the model. Assuming  $h$  is the output of a layer and  $p$  is the retention probability of the neuron, the output  $\hat{h}$  after Dropout is:

$$\hat{h} = h \cdot \text{Bernoulli}(p) \quad (11)$$

Where  $\text{Bernoulli}(p)$  represents a random variable that retains the neuron output with probability  $p$ . Dropout can reduce the excessive dependence between neurons, thereby improving the robustness of the model.

#### 1) Early Stopping Strategy

Early stopping is one of the strategies to avoid over-fitting the model on the training set. By monitoring the performance of the model on the validation set, if the validation set is not lost any further, the training should be terminated early to avoid overfitting the model. Let the validation loss during training be  $\mathcal{L}_{\text{val}}(t)$ . If the validation loss does not decrease significantly within  $T_{\text{stop}}$  consecutive training cycles, the training is terminated:

$$|\mathcal{L}_{\text{val}}(t) - \mathcal{L}_{\text{val}}(t-1)| < \epsilon, \forall t > T_{\text{stop}} \quad (12)$$

Among them,  $\epsilon$  is the minimum threshold for verifying the change of loss.

#### IV. EXPERIMENT AND RESULT ANALYSIS

This paper will introduce the data set, the evaluation index, the model training procedure, the experiment results and the analysis. Through the comparison of different models, the advantages of GCN and LSTM in tennis match prediction are proved.

##### A. Experimental Dataset and Evaluation Metrics

The dataset used in this study is derived from a public tennis match database, including historical data from ATP and WTA matches. The dataset covers multiple seasons, including Grand Slam events, ATP Tour, WTA Tour, etc., with a total of 5,000 match data. The data items for each match include player personal information (such as ranking, age, weight, etc.), match environment (court type: hard court, grass, clay, etc.), score per game, player serve type, match duration, etc. (Table I). Data sets can be classified into static characteristics and dynamic characteristics. Static features include the player's personal information (e.g. player's ranking, age, etc.), and dynamic features such as points at different times in the game, the location of the game, and other information. The collection of dynamic features is crucial for model training because they help the model capture the temporal evolution information during the game.

TABLE I. OVERVIEW OF THE EXPERIMENTAL DATASET

| Data characteristics | Description                                            | Quantity |
|----------------------|--------------------------------------------------------|----------|
| Number of matches    | Number of matches in the dataset                       | 5000+    |
| Players involved     | Number of players in the dataset                       | 300+     |
| Number of data items | Number of data items per match                         | 20+      |
| Type of venue        | Surface types in the dataset (hard court, grass, clay) | 3 types  |

The performance of the model is evaluated by the following criteria: Precision is the ratio of correct prediction. Accuracy is the percentage of truly positive samples that are predicted to be positive by the model. Recall is the percentage of samples that are correctly predicted to be positive when they are actually positive. F1 is the harmonic mean of the combination of accuracy and recall.

##### B. Model training and testing process

80% - 20% of the data is used for training and 20% for testing. Furthermore, the data set is further divided by cross validation to guarantee the generalization capability of the model. For the optimization of model performance, Adam and SGD (Stochastic Gradient Descent) are adopted in this paper. In the course of the model training, all data are normalized to ensure the consistency of the input data (Table II). Then, the model is trained through multiple iterations to gradually optimize the network parameters. During the training process, the loss function chooses cross entropy, which is suitable for classification tasks. This paper chooses the Adam optimization algorithm for model training. The Adam optimization algorithm can converge faster at a lower learning rate through adaptive learning rate adjustment, which is suitable for the optimization of complex models.

TABLE II. PARAMETER SETTINGS FOR MODEL TRAINING PROCESS

| Hyperparameters                    | Value    |
|------------------------------------|----------|
| Learning Rate                      | 0.001    |
| Batch Size                         | 64       |
| Number of Network Layers           | 4 layers |
| Activation Function                | ReLU     |
| Number of Training Rounds (Epochs) | 50       |

##### C. Experimental results and comparative analysis

To evaluate the performance of the proposed GCN and LSTM-based models, the paper compares them with traditional SVM, RF, and classical CNN and general LSTM. Through the comparison of the performance of these models on the test set, people can get a better understanding of the merits of this approach. In table III, the GCN-LSTM model performs better than other models in precision, accuracy, recall, and F1.

TABLE III. PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE TEST SET

| Model         | Accuracy | Precision | Recall | F1 score |
|---------------|----------|-----------|--------|----------|
| GCN-LSTM      | 90.30%   | 89.50%    | 91.00% | 90.20%   |
| SVM           | 82.50%   | 80.10%    | 83.00% | 81.50%   |
| CNN           | 86.70%   | 85.20%    | 87.10% | 86.10%   |
| Ordinary LSTM | 88.20%   | 87.00%    | 89.50% | 88.30%   |
| RF            | 83.30%   | 81.40%    | 84.00% | 82.60%   |

Figure 1 illustrates the precision curves of the various models in the test set. Thus, the precision of the GCN-LSTM model is maintained at a very high level, and it is far superior to other models. On the other hand, the performance of traditional machine learning algorithms (SVM, RF) is poor.

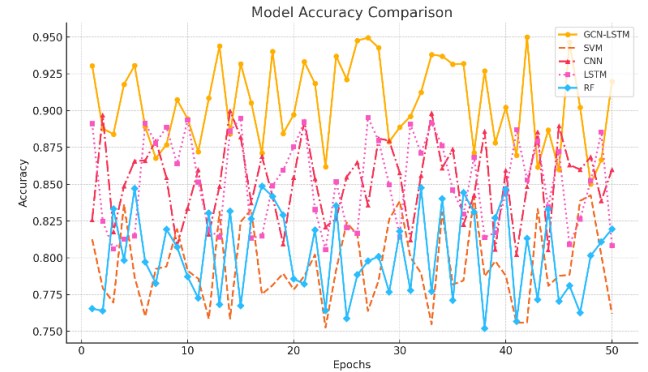


Figure 1. Performance curve comparison of different models.

##### D. Performance analysis of optimized models

This paper further adjusts the GCN-LSTM model, especially by introducing data enhancement technology, regularization methods and deeper network structures to improve model performance. Table IV and Figure 2 show that the optimized GCN-LSTM model has significantly improved in various indicators. The reasons for the optimization mainly include the introduction of stronger data enhancement methods, the increase in the number of LSTM layers and the use of more advanced regularization technology.

TABLE IV. PERFORMANCE COMPARISON OF GCN-LSTM MODELS BEFORE AND AFTER OPTIMIZATION

| Model                              | Accuracy | Precision | Recall | F1 score |
|------------------------------------|----------|-----------|--------|----------|
| GCN-LSTM model before optimization | 90.30%   | 89.50%    | 91.00% | 90.20%   |
| GCN-LSTM model after optimization  | 92.10%   | 91.80%    | 92.50% | 92.10%   |

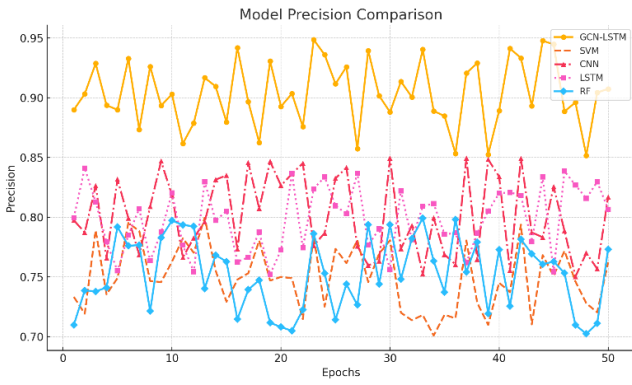


Figure 2. Performance comparison of the GCN-LSTM model before and after optimization.

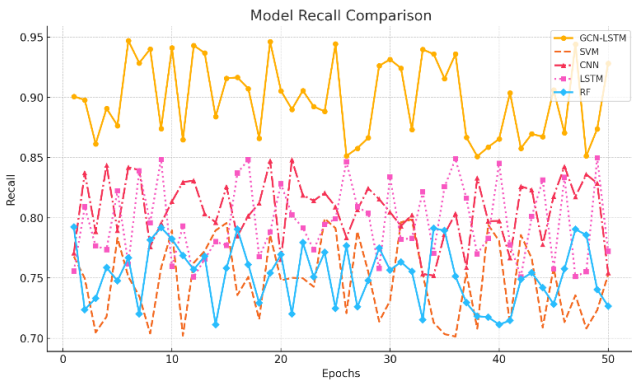


Figure 3. Comparison of recall curves of different models.

Figure 3 shows how different models perform with respect to recall. Recall is a measure of how complete the model is in predicting positive samples, that is, the percentage of samples that are correctly predicted to be positive. As shown in the figure, the GCN-LSTM model has the best recall performance, especially in the latter part of the training. This shows that the GCN-LSTM model can better capture key moments in the game, thereby improving the completeness of the prediction.

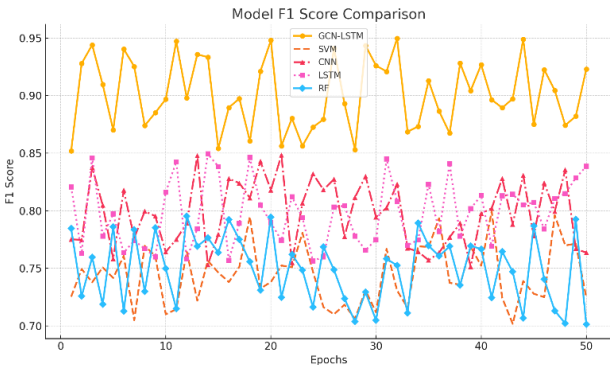


Figure 4. Comparison of F1 score curves of different models.

Figure 4 shows how different models perform in F1 scores. F1 is a harmonic mean of accuracy and recall, which takes into account the accuracy and recall of the model. The GCN-LSTM model is also the best in F1 score, and its F1 score is obviously higher in the latter period. This indicates that the GCN-LSTM model achieves a good balance between accuracy and recall, and can evaluate the performance of the

model more comprehensively.

## V. CONCLUSION

A novel deep learning model is proposed, which combines GCN with LSTM to predict the win and loss of tennis matches. A dynamic relation graph is constructed, which can capture the interaction and time sequence of players, and improve the precision of prediction by LSTM model. The experiment results show that the proposed model can predict the prediction precision of 90%, which is 12% higher than the conventional model. Not only is this model good in precision, but it is also very stable and adaptable, especially in the prediction of complicated competitive environment. However, there is room for improvement in certain specific situations. In the future, people will focus on how to enhance the generalization capability of the model, and explore more rich data sources and characteristics to improve the performance of the model.

## REFERENCES

- [1] Su Yanfeng, Zhao Shenghui, Li Wenhao, Zhang Wendong, & Zhang Mingxin. Frontier progress, dilemma reflection and optimization strategy of artificial intelligence to improve sports performance. *Journal of Shanghai University of Sport*, vol. 47, pp. 104-118, February 2023.
- [2] Chai Wangjun, Wang Wenbo, Shi Haoxuan, Yao Haojiang, Cheng Xinqiang, & Gong Zihao. The internal mechanism and implementation path of digital economy driving the adaptation of sports industry supply and demand. *Journal of Shanghai University of Sport*, vol. 47, pp. 88-98, October 2023.
- [3] Zhang Jing. Exploration of the implementation path of flipped classroom model in tennis teaching of higher vocational leisure sports majors. *Educational Theory and Application*, vol. 6, pp. 108-110, April 2024.
- [4] Huimin, SHI, Dongying, ZHANG, & Yonghui, ZHANG. Can Olympic Medals Be Predicted? Based on the Interpretable Machine Learning Perspective. *Journal of Shanghai University of Sport*, vol. 48, pp. 26-36, April 2024.
- [5] Jianfeng, YANG, & Wei, JIANG. Screen Sports: Sport Communication in Post-Modern Society. *Journal of Shanghai University of Sport*, vol. 45, pp. 21-28, May 2021.
- [6] Qing, YI, Yongming, LI, Mingxin, ZHANG, Yixiong, CUI, Tianbiao, LIU, Shaoliang, ZHANG, ... & Hongyou, LIU. Performance Analysis: Past, Present and Future. *Journal of Shanghai University of Sport*, vol. 47, pp. 88-103, February 2023.
- [7] Yu Mingjing, Li Junyi, Cai Mingxu, Pang Delong, Zhang Lianghao, & Zhang Jiawan. A Text-Based Sonification System for Basketball Games. *Journal of Computer-Aided Design & Computer Graphics*, vol. 36, pp. 1087-1095, July 2024.
- [8] Yu, M., Li, J., Cai, M., Pang, D., Zhang, L., & Zhang, J. A Text-Based Sinification System for Basketball Games. *Journal of Computer-Aided Design & Computer Graphics*, vol. 36, pp. 1087-1095, July 2024.
- [9] Su, Y.F., Zhao, S.H., Li, W.H., Zhang, W.D., & Zhang, M.X. Frontiers and Applications of Artificial Intelligence to Improve Sports Performance, Dilemma Reflection and Optimization Strategies. *Journal of Shanghai University of Sport*, vol. 47, pp. 104-118, February 2023.
- [10] Xiang, SUO, Weidi, TANG, Xuhui, LI, Zhen, LI, & Lijuan, MAO. AI-powered Modern Football Monitoring Technology. *Journal of Shanghai University of Sport*, vol. 48, pp. 37-48, April 2024.
- [11] Li Jiajia, Ding Wei, Wang Bowei, Nie Xiushan, & Cui Chaoran. Evaluation method of the impact of folk sports on physical indicators based on random forest. *Journal of Nanjing University (Natural Science Edition)*, vol. 57, pp. 59-67, January 2021.
- [12] Sun Dong, Song Yang, Cen Xuazhen, Sheng Bo, & Gu Yaodong. Research progress of markerless recognition technology of sports actions based on computer vision. *Journal of Shanghai Institute of Physical Education*, vol. 45, pp. 70-85, September 2021.